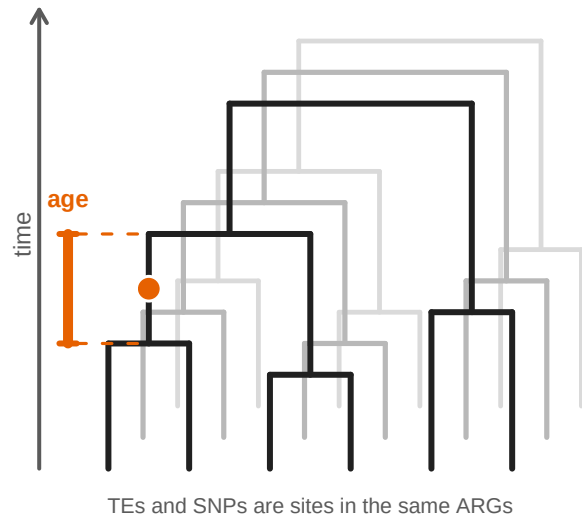


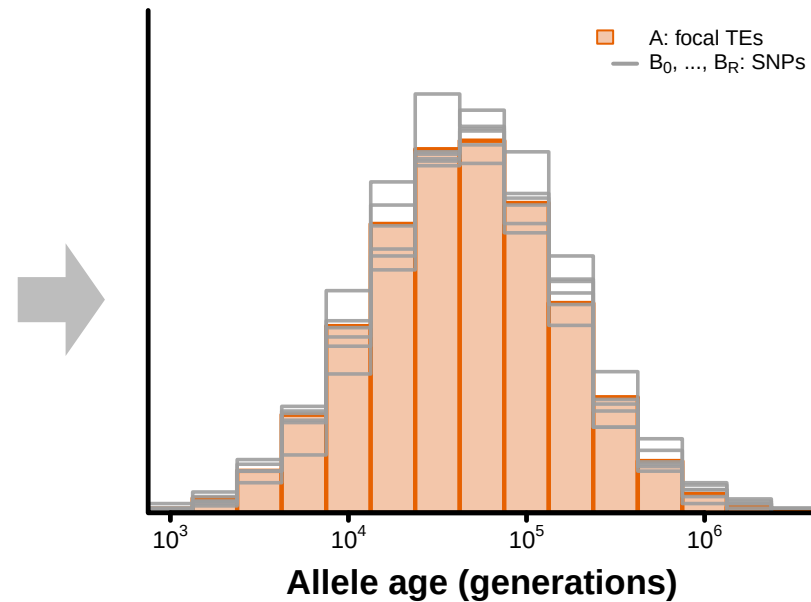
PhiTE: does a TE category's frequency spectrum differ from neutral SNPs of the same age?

1 Posterior ARGs



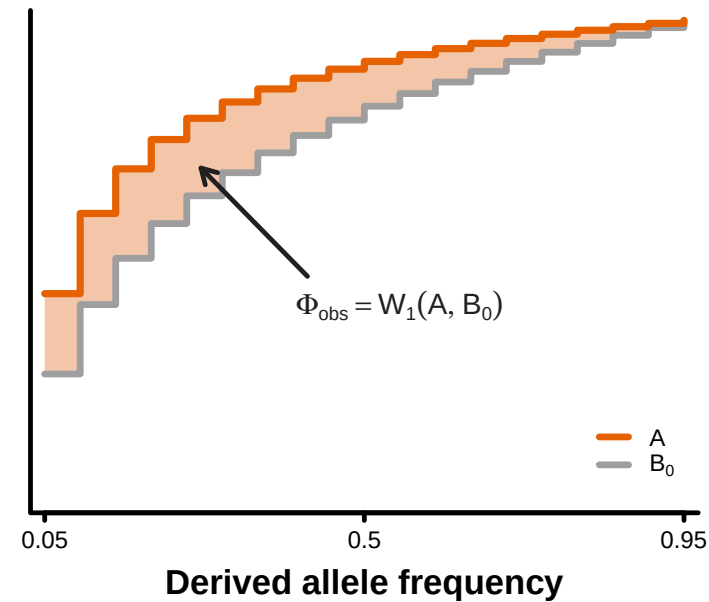
Every site gets an age interval
and $q = P(\text{ALT is derived})$ from
the posterior ARG draws.

2 Age-matched controls



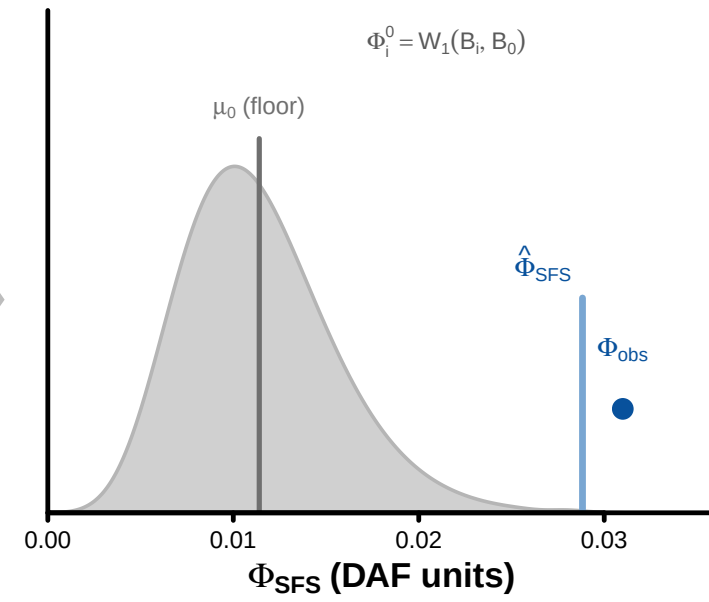
Each control set matches M SNPs
to a bootstrap of A's ages:
500 disjoint sets, frequency-blind.

3 Compare spectra



Project spectra to $m = 20$ and mix
both orientations by q .
 Φ_{SFS} is the area between the CDFs.

4 Calibrate the floor



Null sets give the sampling floor.
Report the add-one P-value and
 $\hat{\Phi}_{\text{SFS}} = \sqrt{\Phi_{\text{obs}}^2 - \mu_0^2}$